# 8

# Basic SQL Query

CSGE602070 - BASIS DATA

# Acknowledgements

This slide is a modification to supplementary slide of
“Database System”, 7th edition, Elmasri/Navathe, 2015: **Chapter 7: More SQL** used in “Basis Data” course in academic years 2018/2019 in the Faculty of Computer Science, Universitas Indonesia.

Additional materials from:
**http://www.postgresql.org/docs/current/static/index.html**

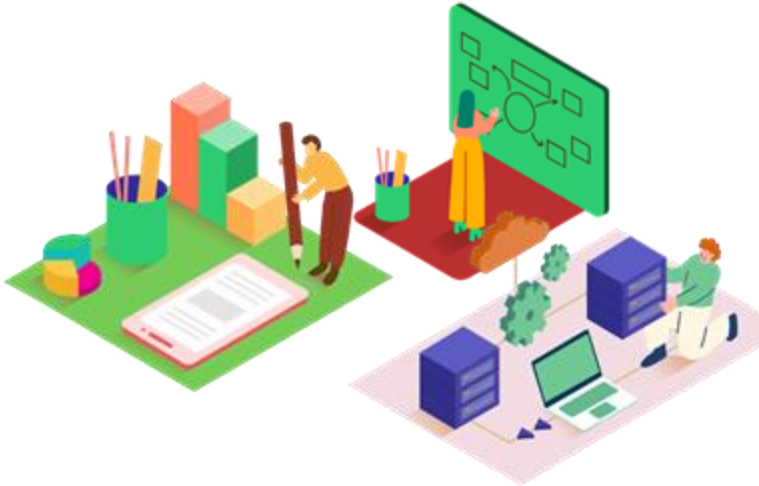
These slides are designed by **Lintang Matahari Hasani, M.Kom.** (lintang.matahari01@cs.ui.ac.id) based on design elements provided by MANYPIXELS under an nonexclusive, worldwide copyright license to download, copy, modify, distribute, perform, and use the assets.

# Outline

1. `SELECT` Statement

2. Aliases and

3. `WHERE` Clause



# Basic Retrieval Queries

SQL allows a table (relation) to have **two or more tuples that are identical** in all their attribute values

```
postgres=# CREATE TABLE STAFF (id VARCHAR(10), nama VARCHAR(100));
CREATE TABLE
postgres=# INSERT INTO STAFF (id, nama) VALUES ('1', 'Naruto Uzumaki');
INSERT 0 1
postgres=# INSERT INTO STAFF (id, nama) VALUES ('1', 'Naruto Uzumaki');
INSERT 0 1
postgres=# SELECT * FROM STAFF;
 id |      nama
----+----------------
 1  | Naruto Uzumaki
 1  | Naruto Uzumaki
(2 rows)
```

- ➔ Different with formal relational model!
- ➔ Use **constraints on DDL**

```
postgres=# DELETE FROM STAFF WHERE id = '1';
DELETE 2
postgres=# ALTER TABLE STAFF ADD PRIMARY KEY (id);
ALTER TABLE
postgres=# \d+ STAFF
                                          Table "company.staff"
 Column |          Type          | Collation | Nullable | Default | Storage  | Compression | Stats target | Description
--------+------------------------+-----------+----------+---------+----------+-------------+--------------+-------------
 id     | character varying(10)  |           | not null |         | extended |             |              |
 nama   | character varying(100) |           |          |         | extended |             |              |
Indexes:
    "staff_pkey" PRIMARY KEY, btree (id)
Access method: heap
```

# **Basic Retrieval Queries** (Cntd.)

Basic statement for retrieving information from a database:

**SELECT** statement

```
SELECT    <attribute list>
FROM      <table list>
WHERE     <condition>;
```

where

- <attribute list> is a list of attribute names whose values are to be retrieved by the query.
- <table list> is a list of the relation names required to process the query.
- <condition> is a conditional (Boolean) expression that identifies the tuples to be retrieved by the query.

---

Logical comparison operators:

| | | | |
|---|---|---|---|
| = | equal to | | |
| | > | more than | |
| < | less than | | >= |
| | more than or equal to | | |
| <= | less than or equal to | | <> |
| | not equal tp | | |

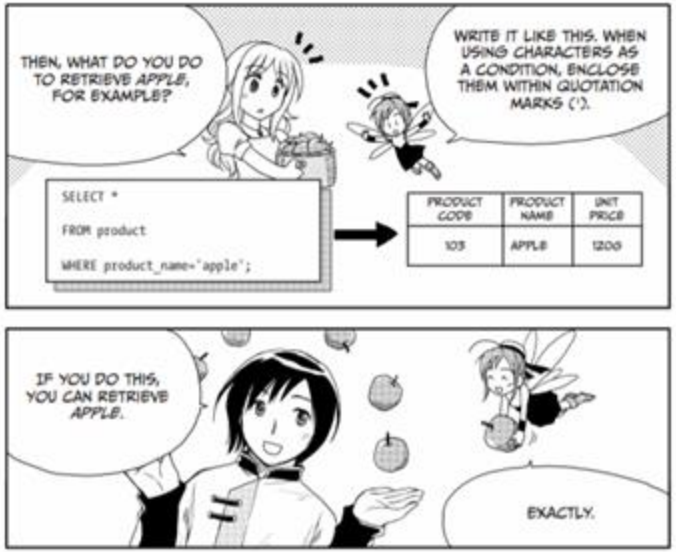



Illustration: Takahashi & Azuma (2014)

# Cartesian Product (Cross Product)

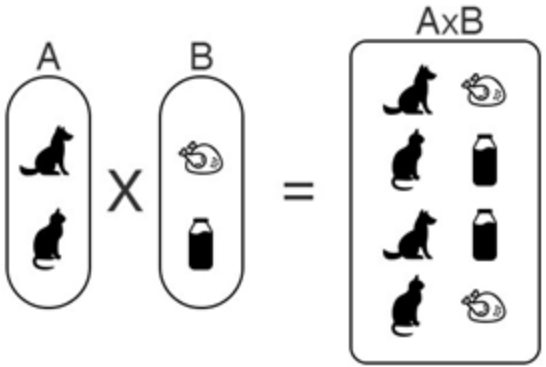
Denoted by ‘x’ operator

Operation that applied on two relations **produces a new element by combining every tuple** from one relation with every tuple from the other relation.

For example:

- ➔ $R(A_1, A_2, \ldots, A_n)$
- ➔ $S(B_1, B_2, \ldots, B_m)$
- ➔ $R \times S = (A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_m)$

If R has |R| tuples and S has |S| tuples, then |R x S| =|R| * |S| tuples




Cartesian Product of Two Sets.

Illustration: Mathtopia

# Cartesian Product Example

**R1**

| A | B | C |
|---|---|---|
| a1 | b1 | c1 |
| a2 | b2 | c2 |
| a3 | b3 | c3 |

**R2**

| X | Y |
|---|---|
| x1 | y1 |
| x2 | y2 |

**R3 = R2 x R1**

| A | B | C | X | Y |
|---|---|---|---|---|
| a1 | b1 | c1 | x1 | y1 |
| a1 | b1 | c1 | x2 | y2 |
| a2 | b2 | c2 | x1 | y1 |
| a2 | b2 | c2 | x2 | y2 |
| a3 | b3 | c3 | x1 | y1 |
| a3 | b3 | c3 | x2 | y2 |

```
postgres=# SELECT * FROM R1,R2;
 a  | b  | c  | x  | y
----+----+----+----+----
 a1 | b1 | c1 | x1 | y1
 a2 | b2 | c2 | x1 | y1
 a3 | b3 | c3 | x1 | y1
 a1 | b1 | c1 | x2 | y2
 a2 | b2 | c2 | x2 | y2
 a3 | b3 | c3 | x2 | y2
(6 rows)
```

# Join

**Combine related tuples** from two relations into single "longer" tuples

General join condition of the form:

```
<condition> AND <condition> AND
<condition> AND … AND <condition>
```

- Tuples that satisfy the join condition will be combined
- Join condition(s) is defined in the WHERE clause

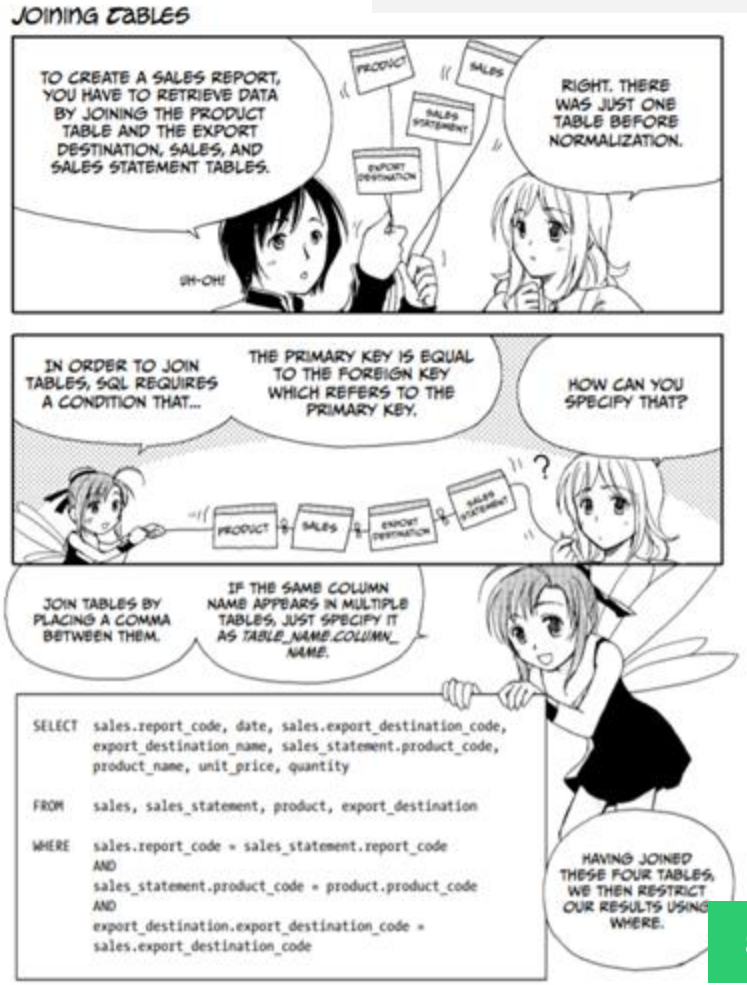



Illustration: Takahashi & Azuma (2014)

# Join Example

**R1**

| A | B | C |
|---|---|---|
| a1 | b1 | c1 |
| a2 | b2 | c2 |
| a3 | b3 | c3 |

**R2**

| X | Y |
|---|---|
| a1 | y1 |
| a1 | y2 |
| a2 | y1 |

```
SELECT * FROM R1, R2 WHERE A = X;
```

| A | B | C | X | Y |
|---|---|---|---|---|
| a1 | b1 | c1 | a1 | y1 |
| a1 | b1 | c1 | a1 | y2 |
| a2 | b2 | c2 | a2 | y1 |

Same data

# Join Example

**Figure 4.3**
Results of SQL queries when applied to the COMPANY database state shown in Figure 3.6. (a) Q0. (b) Q1. (c) Q2. (d) Q8. (e) Q9. (f) Q10. (g) Q1C.

(a)

| Bdate | Address |
|---|---|
| 1965-01-09 | 731Fondren, Houston, TX |

(b)

| Fname | Lname | Address |
|---|---|---|
| John | Smith | 731 Fondren, Houston, TX |
| Franklin | Wong | 638 Voss, Houston, TX |
| Ramesh | Narayan | 975 Fire Oak, Humble, TX |
| Joyce | English | 5631 Rice, Houston, TX |

**Query 0.** Retrieve the birth date and address of the employee(s) whose name is 'John B. Smith'.

```
Q0:   SELECT   Bdate, Address
      FROM     EMPLOYEE
      WHERE    Fname='John' AND Minit='B' AND Lname='Smith';
```

**Query 1.** Retrieve the name and address of all employees who work for the 'Research' department.

```
Q1:   SELECT   Fname, Lname, Address
      FROM     EMPLOYEE, DEPARTMENT
      WHERE    Dname='Research' AND Dnumber=Dno;
```

# Latihan

| EMPLOYEE | | | |
|---|---|---|---|
| emp_no | emp_fname | emp_lname | dept_no |
| 1 | Matthew | Smith | d3 |
| 2 | Ann | Jones | d3 |
| 3 | John | Borrimore | d1 |
| 4 | Andrew | James | d2 |
| 5 | Elisa | Bertoni | d2 |
| 6 | Eike | Hansel | d2 |
| 7 | Sybill | Moser | d1 |

| DEPARTMENT | | |
|---|---|---|
| dept_no | dept_name | location |
| d1 | Developer | Dallas |
| d2 | Tester | Seattle |
| d3 | Marketing | Dallas |

Buatlah SQL statement untuk menjawab pertanyaan berikut ini.

- ➔ Siapakah pegawai yang bekerja sebagai Developer ?
- ➔ Di departemen manakah pegawai bernama depan Matthew bekerja?

# Example

**Figure 4.3**
Results of SQL queries when applied to the COMPANY database state shown in Figure 3.6. (a) Q0. (b) Q1. (c) Q2. (d) Q8. (e) Q9. (f) Q10. (g) Q1C.

(c)

| Pnumber | Dnum | Lname | Address | Bdate |
|---|---|---|---|---|
| 10 | 4 | Wallace | 291Berry, Bellaire, TX | 1941-06-20 |
| 30 | 4 | Wallace | 291Berry, Bellaire, TX | 1941-06-20 |

**Query 2.** For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

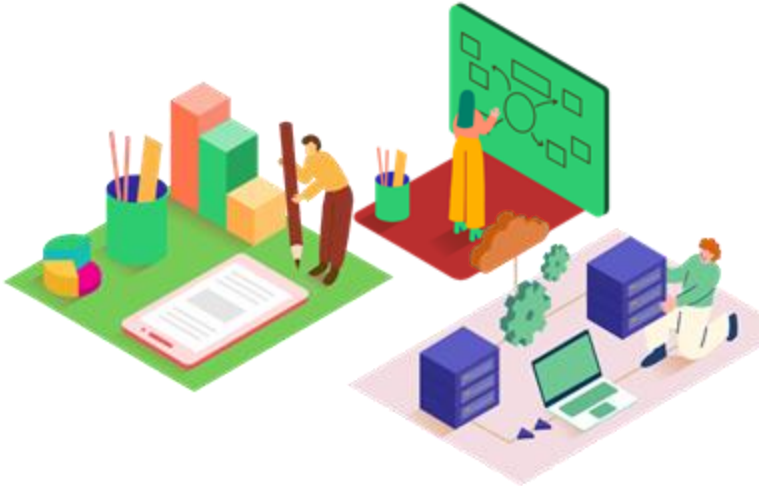
```
Q2:   SELECT   Pnumber, Dnum, Lname, Address, Bdate
      FROM     PROJECT, DEPARTMENT, EMPLOYEE
      WHERE    Dnum=Dnumber AND Mgr_ssn=Ssn AND
               Plocation='Stafford';
```

# Outline

1. `SELECT` Statement

2. Aliases

3. `WHERE` Clause



# Ambiguous Attribute Names

Same name can be used for two (or more) attributes

- ➔ As long as the attributes are in different relations
- ➔ **Must qualify the attribute name with the relation name** to prevent ambiguity

```
Q1A:  SELECT   Fname, EMPLOYEE.Name, Address
      FROM     EMPLOYEE, DEPARTMENT
      WHERE    DEPARTMENT.Name='Research' AND
               DEPARTMENT.Dnumber=EMPLOYEE.Dnumber;
```

# Aliases

Some queries need to refer to the **same relation** twice In this case, aliases are given to the relation name.

**Query 8**:
For each employee, retrieve the employee's name, and the name of his or her immediate supervisor.

```
SELECT E.FNAME, E.LNAME, S.FNAME, S.LNAME
FROM EMPLOYEE E, EMPLOYEE S
WHERE S.SSN = E.SUPER_SSN;
```

**E** and **S** are called aliases, or alternate copies of **EMPLOYEE**

# **Aliases** (Cntd.)

Can also use the **AS** keyword:

```
Q8:   SELECT   E.Fname, E.Lname, S.Fname, S.Lname
      FROM     EMPLOYEE AS E, EMPLOYEE AS S
      WHERE    E.Super_ssn = S.Ssn;
```

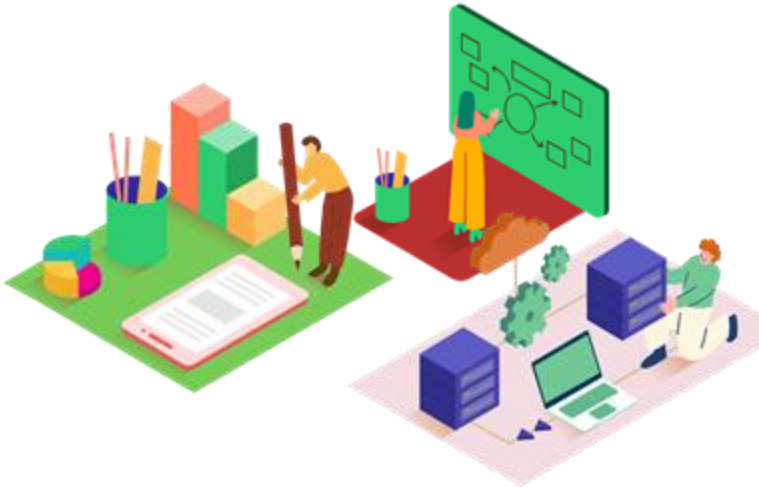
Illustration: https://comicbook.com/gaming/news/marvels-spider-man-ps4-memes/

# Outline

1. `SELECT` Statement

2. Aliases

3. `WHERE` Clause



# Unspecified `WHERE` Clause

A missing WHERE-clause indicates no condition; hence, all tuples of the relations in the FROM-clause are selected.

This is equivalent to the condition WHERE TRUE CROSS PRODUCT: **all posible tuple combination**

**Queries 9 and 10.** Select all EMPLOYEE Ssns (Q9) and all combinations of EMPLOYEE Ssn and DEPARTMENT Dname (Q10) in the database.

```
Q9:   SELECT   Ssn
      FROM     EMPLOYEE;

Q10:  SELECT   Ssn, Dname
      FROM     EMPLOYEE, DEPARTMENT;
```

# Use of Asterisk

Specify an asterisk (*):
Retrieve **all the attribute values** of the selected tuples.

```
Q1C:   SELECT   *
       FROM     EMPLOYEE
       WHERE    Dno = 5;

Q1D:   SELECT   *
       FROM     EMPLOYEE, DEPARTMENT
       WHERE    Dname = 'Research' AND Dno = Dnumber;

Q10A:  SELECT   *
       FROM     EMPLOYEE, DEPARTMENT;
```

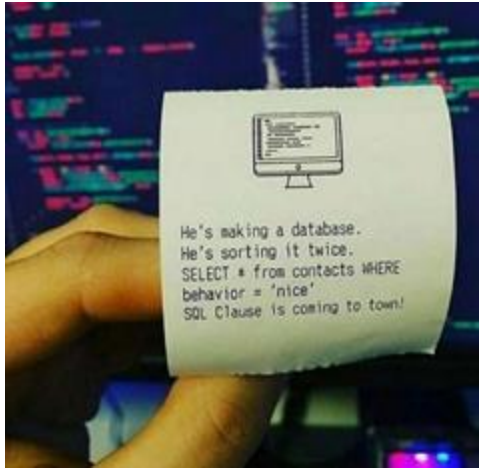



Illustration: https://www.pinterest.com/pin/502221795945959865/

# Tables as Sets in SQL

SQL does not automatically eliminate duplicate tuples in query results.

- ➔ Use the keyword **`DISTINCT`** in the **`SELECT`** clause
- ➔ Only distinct tuples should remain in the result

**Query 11.** Retrieve the salary of every employee (Q11) and all distinct salary values (Q11A).

```
Q11:   SELECT   ALL Salary
       FROM     EMPLOYEE;

Q11A:  SELECT   DISTINCT Salary
       FROM     EMPLOYEE;
```

**Figure 6.4**
Results of additional SQL queries when applied to the COMPANY database state shown in Figure 5.6. (a) Q11. (b) Q11A.

(a)

| Salary |
|---|
| 30000 |
| 40000 |
| 25000 |
| 43000 |
| 38000 |
| 25000 |
| 25000 |
| 55000 |

(b)

| Salary |
|---|
| 30000 |
| 40000 |
| 25000 |
| 43000 |
| 38000 |
| 55000 |

# Tables as Sets in SQL (Cntd.)

## Set operations

`UNION`, `EXCEPT` (difference), `INTERSECT`
Duplicate tuples are eliminated

Corresponding multiset operations: `UNION ALL`, `EXCEPT ALL`, `INTERSECT ALL` – duplicate tuples are not eliminated

(a)

| R |
|---|
| A |
| a1 |
| a2 |
| a2 |
| a3 |

| S |
|---|
| A |
| a1 |
| a2 |
| a4 |
| a5 |

(b)

| T |
|---|
| A |
| a1 |
| a1 |
| a2 |
| a2 |
| a2 |
| a3 |
| a4 |
| a5 |

(c)

| T |
|---|
| A |
| a2 |
| a3 |

(d)

| T |
|---|
| A |
| a1 |
| a2 |

**Figure 6.5**
The results of SQL multiset operations. (a) Two tables, R(A) and S(A).
(b) R(A)UNION ALL S(A).
(c) R(A) EXCEPT ALL S(A).
(d) R(A) INTERSECT ALL S(A).

**Query 4.** Make a list of all project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project.

```
Q4A:  ( SELECT   DISTINCT Pnumber
        FROM     PROJECT, DEPARTMENT, EMPLOYEE
        WHERE    Dnum = Dnumber AND Mgr_ssn = Ssn
                 AND    Lname = 'Smith' )
        UNION
      ( SELECT   DISTINCT Pnumber
        FROM     PROJECT, WORKS_ON, EMPLOYEE
        WHERE    Pnumber = Pno AND Essn = Ssn
                 AND    Lname = 'Smith' );
```

# Substring Pattern Matching and Arithmetic Operators

**`LIKE`** comparison operator

- ➔ Used for string **pattern matching**
- ➔ % replaces an arbitrary number of zero or more characters
- ➔ underscore (_) replaces a single character

Standard arithmetic operators:

- ➔ Addition (+), subtraction (−), multiplication (*), and division (/)

**`BETWEEN`** comparison operator

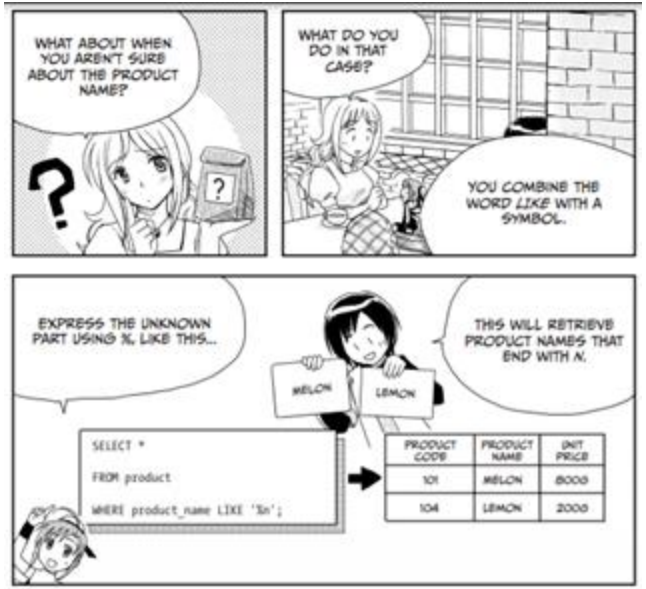



Illustration: Takahashi & Azuma (2014)

# Substring Pattern Matching and Arithmetic Operators (Cntd.)

Example:

```
SELECT Fname, Lname FROM EMPLOYEE WHERE Address LIKE '%Houston,TX%';
```

```
SELECT * FROM EMPLOYEE WHERE (Salary BETWEEN 30000 AND 40000) AND Dno = 5;
```

Equivalent with condition

```
((Salary) >= 30000) AND (Salary <= 40000))
```

# Ordering of Query Results

Use ORDER BY clause

- ➔ Keyword DESC to see result in a descending order of values
- ➔ Keyword ASC to specify ascending order explicitly

```
ORDER BY D.Dname DESC, E.Lname ASC, E.Fname ASC;
```

```
SELECT      <attribute list>
FROM        <table list>
[ WHERE     <condition> ]
[ ORDER BY  <attribute list> ];
```

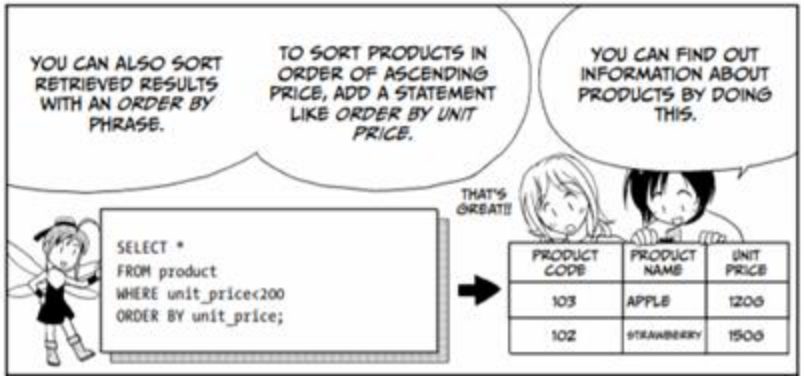



Illustration: Takahashi & Azuma (2014)

# Exercise

**Figure 5.6**
One possible database state for the COMPANY relational database schema.

**EMPLOYEE**

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|
| John | B | Smith | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M | 30000 | 333445555 | 5 |
| Franklin | T | Wong | 333445555 | 1955-12-08 | 638 Voss, Houston, TX | M | 40000 | 888665555 | 5 |
| Alicia | J | Zelaya | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F | 25000 | 987654321 | 4 |
| Jennifer | S | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F | 43000 | 888665555 | 4 |
| Ramesh | K | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M | 38000 | 333445555 | 5 |
| Joyce | A | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX | F | 25000 | 333445555 | 5 |
| Ahmad | V | Jabbar | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M | 25000 | 987654321 | 4 |
| James | E | Borg | 888665555 | 1937-11-10 | 450 Stone, Houston, TX | M | 55000 | NULL | 1 |

**DEPARTMENT**

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|
| Research | 5 | 333445555 | 1988-05-22 |
| Administration | 4 | 987654321 | 1995-01-01 |
| Headquarters | 1 | 888665555 | 1981-06-19 |

**DEPT_LOCATIONS**

| Dnumber | Dlocation |
|---|---|
| 1 | Houston |
| 4 | Stafford |
| 5 | Bellaire |
| 5 | Sugarland |
| 5 | Houston |

**WORKS_ON**

| Essn | Pno | Hours |
|---|---|---|
| 123456789 | 1 | 32.5 |
| 123456789 | 2 | 7.5 |
| 666884444 | 3 | 40.0 |
| 453453453 | 1 | 20.0 |
| 453453453 | 2 | 20.0 |
| 333445555 | 2 | 10.0 |
| 333445555 | 3 | 10.0 |
| 333445555 | 10 | 10.0 |
| 333445555 | 20 | 10.0 |
| 999887777 | 30 | 30.0 |
| 999887777 | 10 | 10.0 |
| 987987987 | 10 | 35.0 |
| 987987987 | 30 | 5.0 |
| 987654321 | 30 | 20.0 |
| 987654321 | 20 | 15.0 |
| 888665555 | 20 | NULL |

**PROJECT**

| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|
| ProductX | 1 | Bellaire | 5 |
| ProductY | 2 | Sugarland | 5 |
| ProductZ | 3 | Houston | 5 |
| Computerization | 10 | Stafford | 4 |
| Reorganization | 20 | Houston | 1 |
| Newbenefits | 30 | Stafford | 4 |

**DEPENDENT**

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|
| 333445555 | Alice | F | 1986-04-05 | Daughter |
| 333445555 | Theodore | M | 1983-10-25 | Son |
| 333445555 | Joy | F | 1958-05-03 | Spouse |
| 987654321 | Abner | M | 1942-02-28 | Spouse |
| 123456789 | Michael | M | 1988-01-04 | Son |
| 123456789 | Alice | F | 1988-12-30 | Daughter |
| 123456789 | Elizabeth | F | 1967-05-05 | Spouse |

1. Retrieve the names of all employees in dept. 5 who work more than 10 hours per week on the ProductX project
2. List the names of all employees who have a dependent with the same first name as themselves
3. Find the names of all employees who directly supervised by 'Franklin Wong''

# Q&A